

Requirements

Functional requirements:

- FR1: The system shall be able to ingest logs of different formats.
- FR2: The system shall allow data from both log-files based sources as well as event streams.
- FR3: The system shall be able to normalise the different log data into a unified format.
- FR4: The system shall save all normalised log records into a database.
- FR5: The system shall automatically detect anomalies in the ingested log data.
- FR6: The system shall report identified anomalies to the user via a dashboard interface.
- FR7: The system shall display which ATMs are currently non-functional or have warnings.
- FR8: The system shall generate a probable root cause for detected anomalies.
- FR9: The system shall identify anomalous patterns in log data across different channels.
- FR10: The system should use identified patterns to predict potential future anomalies.
- FR11: The system should display recommended next actions for anomalies.
- FR12: The system should allow users to filter data and alerts by specific ATM-IDs.
- FR13: The system should provide a configurable data retention period, after which records are purged.

Non-functional requirements:

- NFR1: The system shall enforce role-based access control, distinguishing between standard user and administrator privilege levels.
- NFR2: The system shall handle malformed or incomplete log records without crashing.
- NFR3: The system shall support concurrent ingestion from multiple log sources without data loss or database corruption.
- NFR4: The system shall preserve unresolved anomaly records regardless of data retention period.

- NFR5: The system shall provide a usable interface for both technical and non-technical users.
- NFR6: The source code shall be well commented, version-controlled and structured to support future maintenance and development.
- NFR7: The system should be extensible, allowing new log source types to be added without modification to the core log ingestion pipeline.

Personas

Persona 1

Role: Security

Name: Bob Security

Age: 53

Background: Long term employee with extensive cybersecurity experience

Goal: Want to maintain and ensure that systems are operating securely with no malicious breaches or denials of service.

Interest: Frequently stay up to date on ongoing logs and be informed of anomalies within the systems that may indicate ongoing security breaches.

Pain Points:

- Ongoing system data that needs to be observed is disjointed, unformatted and heterogeneous, making it difficult to teach to new employees and time consuming to parse.
- All the leg work has to be done on their part to make deductions of trends and anomalies.

Key Features Needed:

- Central space that displays system logs across systems in a more digestible, homogenised form.
- Anomaly indicators, predictions and comments

Persona 2

Role: Management

Name: Steven Smith

Age: 45

Background: New to the job, Undergrad in Business Management (not very tech savvy)

Goals:

- Understand system performance through clear dashboards and insight summaries.
- Make data-driven product and operational decisions based on insights from log analysis.
- Identify trends or recurring issues that may affect system reliability or customer experience

Pain Points:

- Requires easy to understand, technical jargon free, analysis and summaries.
 - High level visual summaries not technical data

Key Features Needed:

- Effective predicting anomaly detection
- Clear dashboards and visuals
- Simple explanations of anomalies detected

- High level reports for decision making

Persona 3

Role: Data Analyst

Name: Linus Torvalds

Age: 35

Background: Experienced data analyst with a masters in data science. Comfortable working with large datasets, statistical analysis and data pipelines. Responsible for analysing system logs and identifying patterns that indicate anomalies

Goals:

- Access structures log data that easily cleaned and analysed
- Identify patterns, correlations and anomalies across multiple log types / sources
- Support engineering and management teams by making meaningful insights from the data

Pain Points:

- Logs from multiple sources are often times inconsistent or unstructured
- Data cleaning can take a long time before analysis can start
- Difficult to correlate events across different systems without shared identifiers

Key Features Needed:

- Structured and consistent log formats
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Persona 4

Role: Software Engineer

Name: John Davis

Age: 50

Background: Worked in the software engineer department of multiple companies and very familiar with logging systems.

Goal:

- To maintain, improve and polish functioning systems.
- Responsible for bug fixing.

How They Use System:

- Utilise logs/anomalies to track/find the source of bugs and flaws within the system.
- Utilise logs/anomalies to gain information about larger data sets and issues within the system.

Pain Points:

- Spending time trying to find the source of errors in logs manually, sometimes wading through large amounts of lines of data.
- Having to identify trends in data themselves that might indicate long term inefficiency.

Key Features Needed:

- Summary of source/cause of issues.
- Trend reports.
- Flag up system of anomalies within logs.

Persona 5

Role:

ATM Engineer

Name:

Tony White

Age: 32

Background: Competent ATM Engineer who's worked in the field for a few years

Goals:

- Ensure that ATMs are always working
- Understand system health trends
- Make real-time decisions based on data

Pain Points:

- Current monitoring tools are too detailed
- Hard to get high level summaries
- Difficult to measure long-term trends

How They Use The System:

- View summary dashboards
- Monitor uptime and anomaly trends

Key Features Needed:

- Easy-to-understand
- Trend Reports

User Stories and Epics

Epic 1 - Log Ingestion

- As a Product Owner I want formatted logs so that i can have a consistent structure for system analysis within the company
 - Take logs from different system
 - Turn into a consistent format
 - Store recent logs in a database engine
 - Don't maintain extensive history (security reasons)

Epic 2 - Anomaly Detection Engine

- As a software engineer I want anomalies to be flagged so that I can identify problems
 - Detect anomalous data in logs
 - Anomaly is reported to user
- As a ... I want effective predictive anomaly detection so that i can predict when problems will occur before they happen
 - Have historical log data
 - Identify patterns in log data
 - Use patterns to predict future anomaly occurrences

Epic 2 - Anomaly Analysis

- As I want simple explanations of anomalies detected so that I can understand what needs to be fixed
 - Generate explanation of anomaly
 - Encapsulate numerous different reports of the same anomaly into 1 explanation
- As a ... I want patterns in data to be tracked so that proper fixes can be made
 - Certain anomalies specifically pertain to patterns in the data

Epic 3 - Visualisation/High Level Interpreting

- As a ... I want a view of charts, timelines and summaries so that I have a clear visual overview of relevant information
 - Generate views of datasets
 - Display views on dashboard
- As a ... I want to only see information that is relevant to me, so that I can focus on
 - Split anomalies into different categories
 - Allow filtering of specific log types
- As a ... I want anomalies that arrive at the same time grouped together so that there is reduced clutter
 - Group similar anomalies based on timestamp

Epic 4 - Admin Controls/Security

- As a system admin I want to be able to allocate access to certain users so that security is maintained
 - Assign access roles/levels to individual users
 - Differentiate what anomalies and logs each role can access